

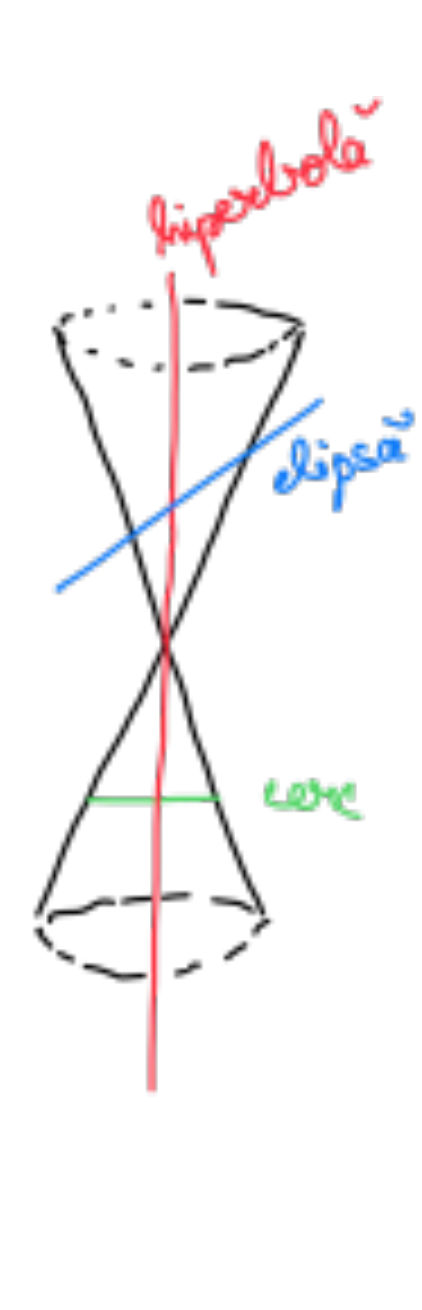
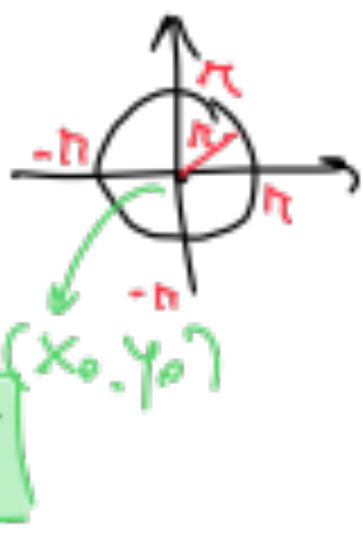
Recapitulare geometrie

A Geometrie în plan

1. Cercul

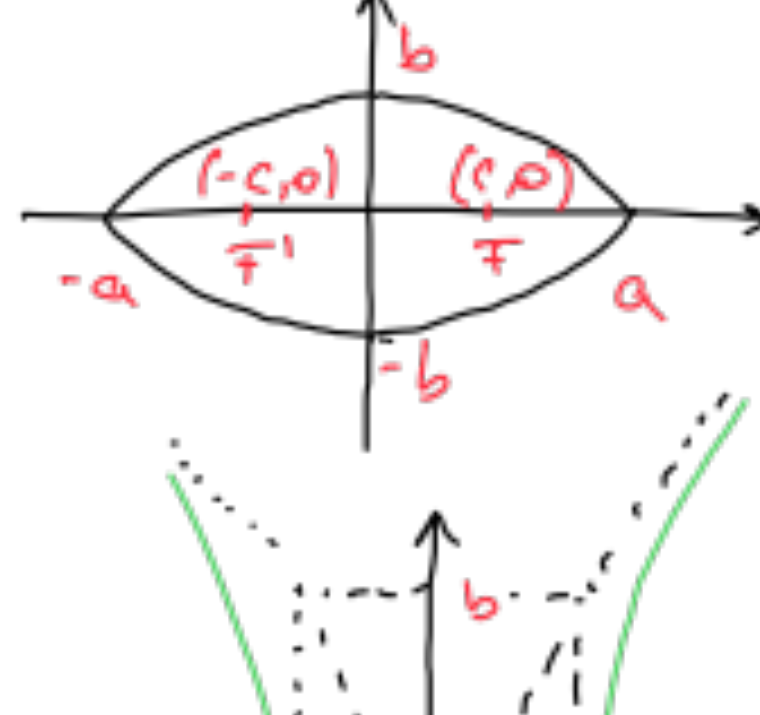
$$x^2 + y^2 = r^2$$

$$(x - x_0)^2 + (y - y_0)^2 = r^2$$



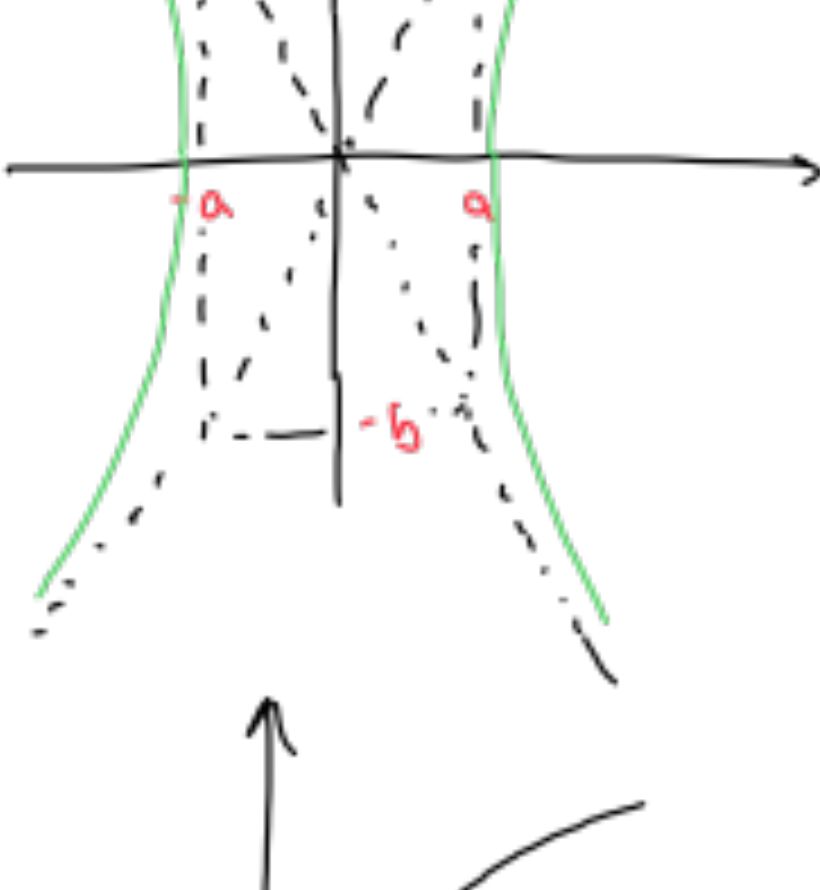
2. Elipsa

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$



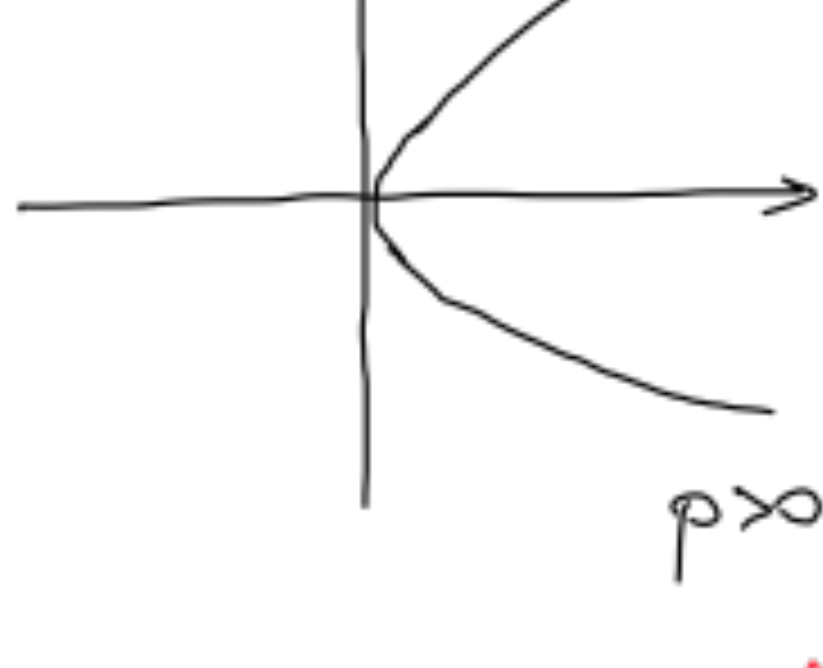
3. Hiperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$



4. Parabola

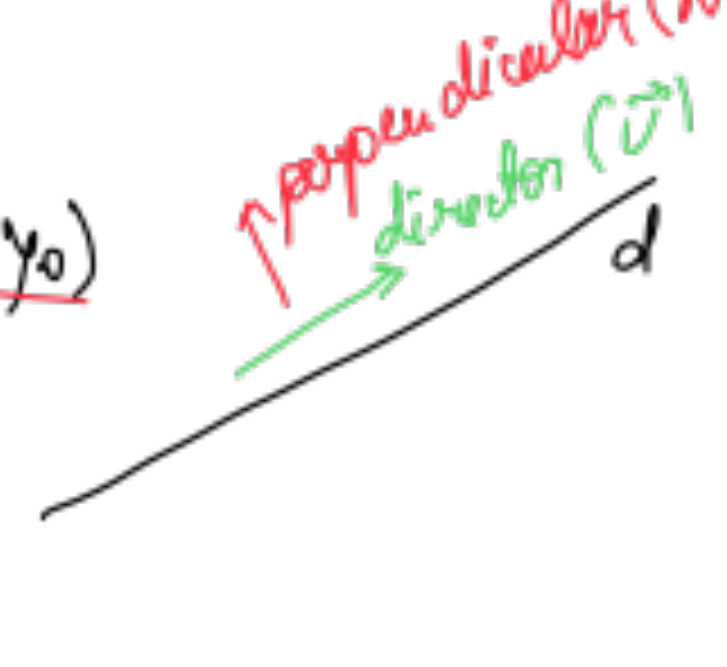
$$y^2 = 2px$$



B Ecuații dreptei în plan

1. Ec. dr. care trece prin $M_0(x_0, y_0)$ și are vect. dir. $\vec{V} = a\vec{i} + b\vec{j}$

$$\frac{x - x_0}{a} = \frac{y - y_0}{b}$$



2. Ec. dr. care trece prin $M_0(x_0, y_0)$ și are vect. norm. $\vec{N} = (a, b) = a\vec{i} + b\vec{j}$

$$a(x - x_0) + b(y - y_0) = 0$$

3. Ec. dr. care trece prin $M_0(x_0, y_0)$ și are panta m

$$y - y_0 = m(x - x_0)$$

4. Ec. dr. care trece prin două puncte $M_1(x_1, y_1)$ și $M_2(x_2, y_2)$

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

$$d: \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

5. Ec. canonică a unei drepte

$$ax + by + c = 0$$

Exerciții

1. Să se scrie ec. dreptei dacă se cunosc un punct $A(1, -1)$ al dr. și

a) vect. dir. $\vec{V} = (-1, 2)$

b) pct. $B(2, 0)$ al dr.

c) vect. norm. $\vec{N} = (1, 3)$

d) panta $m = 4$

a) $\frac{x-1}{-1} = \frac{y+1}{2} \Rightarrow x-1 = \frac{y+1}{2} \Rightarrow -2x+2 = y+1$

$$y = -2x+1$$

b) $\frac{x-1}{2-1} = \frac{y+1}{0-1} \Rightarrow x-1 = y+1 \Rightarrow y = x-2$

c) $1(x-1) + 3(y+1) = 0 \Rightarrow x-1+3y+3=0$

$$3y = -x-2 \Rightarrow y = -\frac{x+2}{3}$$

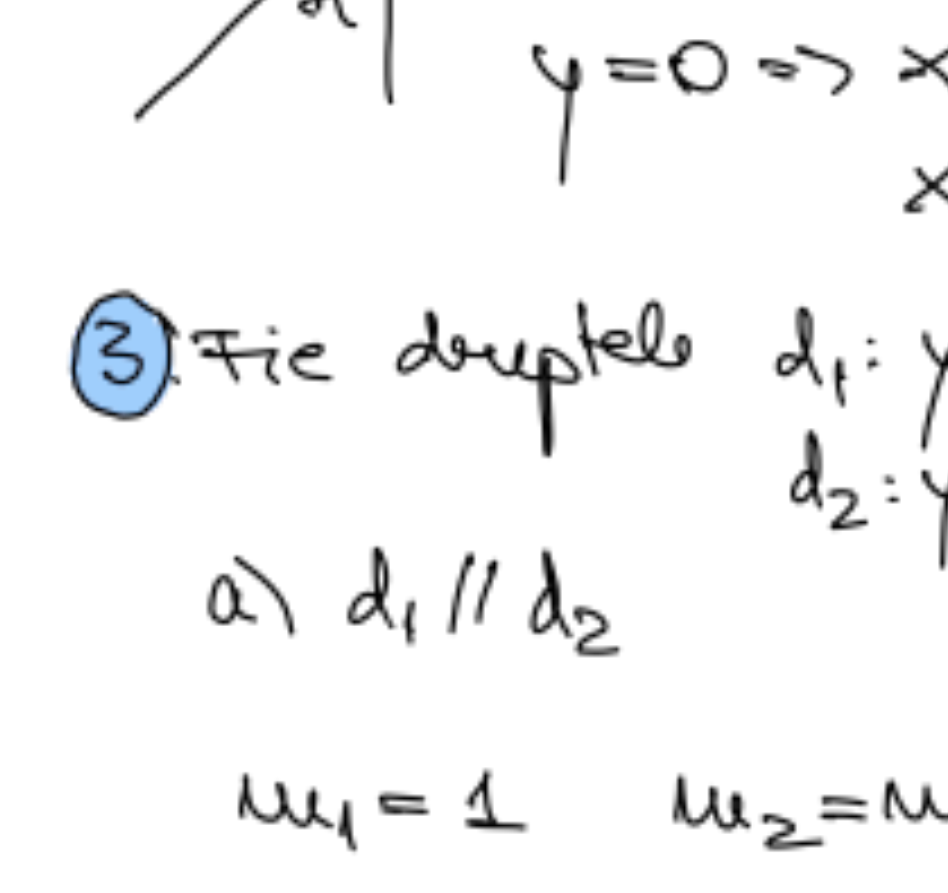
d) $y+1 = 4(x-1) \Rightarrow y = 4x-5$

2. Să se determine pct. de intersecție al dr.

$$x - y - 2 = 0 \text{ și } x + y - 6 = 0$$

$$\begin{cases} x - y - 2 = 0 \\ x + y - 6 = 0 \end{cases} \Rightarrow \begin{cases} x = y + 2 \\ x + y - 6 = 0 \end{cases} \Rightarrow$$

$$\Rightarrow y+2+y-6=0 \Rightarrow 2y=4 \Rightarrow y=2 \Rightarrow x=4 \Rightarrow M(4, 2)$$



$$x=0 \Rightarrow 0 - y - 2 = 0 \Rightarrow y = -2 \Rightarrow A(0, -2)$$

$$\Rightarrow 0 + y - 6 = 0 \Rightarrow y = 6 \Rightarrow D(0, 6)$$

$$y=0 \Rightarrow x - 2 = 0 \Rightarrow x = 2 \Rightarrow C(2, 0)$$

$$x + 0 - 6 = 0 \Rightarrow x = 6 \Rightarrow B(6, 0)$$

3. Fie dreptele $d_1: y = x+2$ determ. m o. f.

$$d_2: y = mx+1$$

a) $d_1 \parallel d_2$

b) $d_1 \perp d_2$

$$m_1 = 1 \quad m_2 = m$$

a) $m_1 = m_2 \Rightarrow m = 1$

b) $m_1 \cdot m_2 = -1 \Rightarrow m = -\frac{1}{m_1} = -1 \Rightarrow m = -1$

4. Să se scrie ec. dr. care trece prin $A(3, -1)$ și formează cu axa Ox unghi de $\frac{\pi}{4}$

$$m = \tan \alpha = \tan \frac{\pi}{4} = 1$$

$$d: y+1 = 1(x-3) \Rightarrow y = x-3-1 \Rightarrow y = x-4$$

5. Verificați coliniaritatea punctelor $A(1, 2); B(2, -2); C(3, -3)$

$$\begin{vmatrix} 1 & 2 & 1 \\ 2 & -2 & 1 \\ 3 & -3 & 1 \end{vmatrix} = -\frac{1}{2} + \frac{1}{6} + \frac{3}{2} + 3 - 4 = 0$$

$$\Rightarrow A_6 = 0 \Rightarrow \text{pct. coliniare}$$

6. Să se scrie ec. dr. care trece prin $A(0, -4)$ și este egal depărtată de pct. $B(-7, 2); C(-9, -6)$

$$\text{fie } M \text{ mij } BC \Rightarrow x_M = \frac{-7-9}{2} = -8 \Rightarrow M(-8, -2)$$

$$y_M = \frac{2-6}{2} = -2$$

$$AM: \begin{vmatrix} x & y & 1 \\ 0 & -4 & 1 \\ -8 & -2 & 1 \end{vmatrix} = 0 \Rightarrow -4x - 8y - 32 + 2x = 0$$

$$8y = -2x - 32$$

$$4y = -x - 16 \Rightarrow y = -\frac{x}{4} - 4$$

$$\Rightarrow m_{AM} = -\frac{1}{4}$$

$$BC: \begin{vmatrix} x & y & 1 \\ -7 & 2 & 1 \\ -9 & -6 & 1 \end{vmatrix} = 0 \Rightarrow 2x - 9y + 42 + 18 + 6x + 9y = 0$$

$$8x - 2y + 60 = 0$$

$$4x - y + 30 = 0$$

$$y = 4x + 30 \Rightarrow m_{BC} = 4$$

$$m_{BC} \cdot m_{AM} = 4 \cdot (-\frac{1}{4}) = -1 \Rightarrow BC \perp AM$$

$$AM \text{ mediatorea lui } BC \Rightarrow \text{ec. } AM: y = -\frac{x}{4} - 4$$

7. Să se det. coord. proiectiei pct. $A(2, 3)$ pe dr.

$$2x - y + 2 = 0 \Rightarrow y = 2x + 2$$

(V1) $m_d = 2$

$$m_{d_1} \cdot m_d = -1 \Rightarrow m_{d_1} = -\frac{1}{2}$$

$$d_1: y - y_A = m_{d_1}(x - x_A) \Rightarrow y - 3 = -\frac{1}{2}(x - 2) \Rightarrow$$

$$\Rightarrow d_1: y = -\frac{1}{2}x + 4$$

$$d_1 \cap d = \text{proiect. lui } A = A'$$

$$2x + 2 = -\frac{1}{2}x + 4 \Rightarrow \frac{5}{2}x = 2 \Rightarrow x = \frac{4}{5}$$

$$y = \frac{8}{5} + 2 = \frac{18}{5} \Rightarrow A'(\frac{4}{5}, \frac{18}{5})$$

(V2) $d: 2x - y + 2 = 0 \Rightarrow$ vect. norm. $\vec{N} = (2, -1)$

$$\vec{N} \text{ vect. dir. pt } d_1 \Rightarrow \frac{x-2}{2} = \frac{y-3}{-1} \Rightarrow$$

$$\Rightarrow d_1: -x+2 = 2y-6 \Rightarrow d_1: 2y = -x+8 \Rightarrow$$

$$d_1: y = -\frac{1}{2}x + 4$$

$$\{P\} = d_1 \cap d \Rightarrow \begin{cases} y = -\frac{1}{2}x + 4 \\ 2x - y + 2 = 0 \end{cases}$$

8. Să se afle coord. pct. B simetric pct. $A(2, 4)$ față de dr. $x - y - 2 = 0$

$$\vec{N} = (1, -1), \text{ dar } \vec{N} \text{ vect. dir. pt } d' \Rightarrow$$

$$\Rightarrow \frac{x-2}{1} = \frac{y-4}{-1} \Rightarrow y-4 = 4-x \Rightarrow y = -x+8$$

$$\{P\} = d' \cap d \Rightarrow \begin{cases} x - y - 2 = 0 \\ y = -x + 8 \end{cases} \Rightarrow x + x - 6 - 2 = 0$$

$$x = 4 \Rightarrow y = 2 \Rightarrow P(4, 2)$$

$$x_P = \frac{x_A + x_B}{2} \Rightarrow x_B = 2x_P - x_A = 8 - 2 \Rightarrow x_B = 6$$

$$y_B = 2y_P - y_A = 4 - 4 \Rightarrow y_B = 0 \Rightarrow B(6, 0)$$

9. Să se scrie ec. simetricii dr. $x - y - 1 = 0$ în raport cu dr. $y = \frac{1}{2}x$

$$d: y = \frac{1}{2}x \Rightarrow m_d = \frac{1}{2}$$

$$d_1: y = x - 1 \Rightarrow m_{d_1} = 1$$

$d_2?$

$$d \cap d_1 = \{P\} \Rightarrow \begin{cases} y = \frac{1}{2}x \\ y = x - 1 \end{cases} \Rightarrow \frac{1}{2}x = x - 1 \Rightarrow \frac{1}{2}x = 1 \Rightarrow x = 2 \Rightarrow y = 1 \Rightarrow P(2, 1)$$

$$\text{Fie } x_1 = 1 \text{ și } x \in d_1 \Rightarrow y_1 = 0 \Rightarrow A_1(1, 0)$$

$$\text{Fie } A_2 \text{ proiectia lui } A_1 \text{ pe dr. } d_2$$

$$A_1 A_2 \perp d \Rightarrow m_{A_1 A_2} \cdot m_d = -1 \Rightarrow m_{A_1 A_2} = -2$$

$$A_1 A_2: x - 1 = -2(y - 0) \Rightarrow A_1 A_2: x - 1 = -2y \Rightarrow A_1 A_2: x + 2y - 1 = 0$$

$$\text{Fie } A_1 A_2 \cap d = \{P\} \text{ atunci } M \text{ mij } A_1 A_2$$

$$\downarrow$$

$$\begin{cases} y = \frac{1}{2}x \\ x + 2y - 1 = 0 \end{cases} \Rightarrow x + x - 1 = 0 \Rightarrow x = \frac{1}{2}$$

$$\Rightarrow y = \frac{1}{4} \Rightarrow M(\frac{1}{2}, \frac{1}{4})$$

$$M \text{ mij } A_1 A_2 \Rightarrow x_M = \frac{x_{A_1} + x_{A_2}}{2} \Rightarrow x_{A_2} = 2x_M - x_{A_1} = 1 - 1 = 0$$

$$y_M = \frac{y_{A_1} + y_{A_2}}{2} \Rightarrow y_{A_2} = 2y_M - y_{A_1} = \frac{1}{2} - 0 = \frac{1}{2}$$

$$\Rightarrow A_2(0, \frac{1}{2}) \in d_2$$

$$d_2 = A_1 P: \begin{vmatrix} x & y & 1 \\ 2 & 1 & 1 \\ 0 & \frac{1}{2} & 1 \end{vmatrix} = 0 \Rightarrow x + 0 + 1 - \frac{1}{2}x - 2y - 0 = 0$$

$$d_2: \frac{1}{2}x - 2y + 1 = 0$$

